

Indigenous Environmental Data

Science

Teacher Guide



About Us

These slides are a collaboration between:

[Victoria Thomson](#), an Albert Einstein Distinguished Educator Fellow with USGS and high school science teacher, Christine Martin, PhD Candidate at Montana State University, and Rachel Torres, a postdoctoral scholar at Cal Poly Humboldt

The larger project and coding activities were created through a collaboration between Georgia Smies, an instructor at Salish Kootenai College, Sean Dorr, Jason McLachlan, and students at the University of Notre Dame. It is funded by the Alfred P. Sloan Foundation.

Our goal is to increase equitable pathways in earth and environmental data science.



Time and Structure of Indigenous Environmental Data Science



Piloted at Salish Kootenai College in Montana Winter Quarter 2024
Course: Intro to R Statistics in the Wildlife and Fisheries Dept.

Task: Complete Modules 1 - 3



Each Module requires ~3 weeks (6 hr class time) + outside class time

Frequency of Meetings:

Meet 1x per week for 2 hours

Total of 10 weeks



Class Format:

First hour of class: direct instruction/discussion



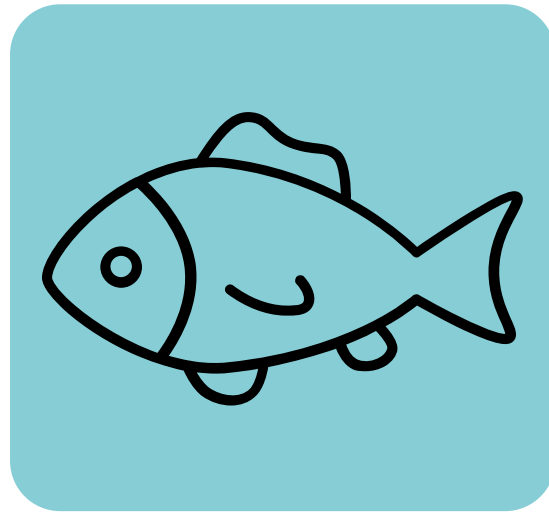
Background and Purpose



- Why are we doing this?
 - To introduce you to a fictional tribe K'avi so you can analyze data relevant to indigenous communities
 - Code can appear scary; but the sooner you play with it, you will see it's not so bad! It can be really powerful for you to know and understand how to use this programming language
 - To empower you to ask questions that are relevant to your community and identify how to share the information learned from your questions
- What is data science?
 - Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from data (source: [datacamp](#))
- What is R?
 - R is a programming language commonly used to analyze large sets of data



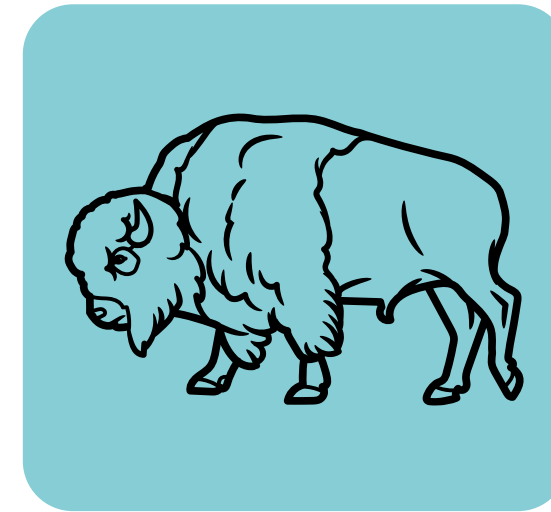
Indigenous Environmental Data Science Modules



Module 1: Mercury
in Fish



Module 2: Water
Quality



Module 3: Bison
Reintroduction



Module 4: Forestry
Management



[Github](#)



GitHub is like a big online notebook where people can save and share their coding projects.

Imagine you're working on a school project with friends. Instead of passing a notebook back and forth, GitHub lets you all work on it at the same time and saves every change.

It helps people:

- ✓ Store their code safely
- ✓ Work together on projects
- ✓ Keep track of all changes

Think of it as Google Docs, but for coding! 🚀



[Environmental Indigenous Data Science](#)
[Code on Github](#)

Choose Your Path



[R and R studio](#)
[Free, but requires](#)
[downloading software](#)

[Best for:](#)
[Laptops with memory](#)
[Computer labs](#)



[binder](#)
[Free, cloud-based service](#)
[to run notebooks](#)

[Best for:](#)
[Any environment](#)



Google Colab
Free, cloud-based
service
Connects to Google
Classroom?
Works with
chromebooks?

Best for:
Any environment



Day

1

Lesson 0: Getting Started in R

Goals:

- Download and install R and R studio
- Identify the 4 main windows of the R interface
- Create a project connected to a Github repository
- Identify:
 - Code chunk
 - Symbols for beginning and ending a code chunk
 - Comments
- Clear the environment
- Run a chunk of code
- Install and load packages
 - tidyverse
 - ggplot2



Getting Started—download R and R Studio

Step 1: Install R

1. Go to the official R download page:
👉 <https://cran.r-project.org/>
1. Click "**Download R for Windows**".
2. Click "**base**", then download the latest **R-x.x.x-win.exe** file.
3. Run the installer and follow these steps:
 - Click **Next** until you reach the installation location.
 - Use the **default settings** unless you have a reason to change them.
 - Finish the installation.

Step 2: Install RStudio

1. Go to the RStudio download page:
👉 <https://posit.co/download/rstudio-desktop/>
1. Download **RStudio Desktop (Free)** for Windows.
2. Run the installer and follow the instructions (keep default settings).
3. Once installed, open **RStudio**. It should automatically detect R.

Identify the 4 windows in R

1. Source Editor
2. Environments
3. Console
4. Output

The screenshot displays the RStudio interface with four main windows identified by red labels:

- Source Editor:** Located at the top left, it shows the R script file `ggplot2.R` with the following code:

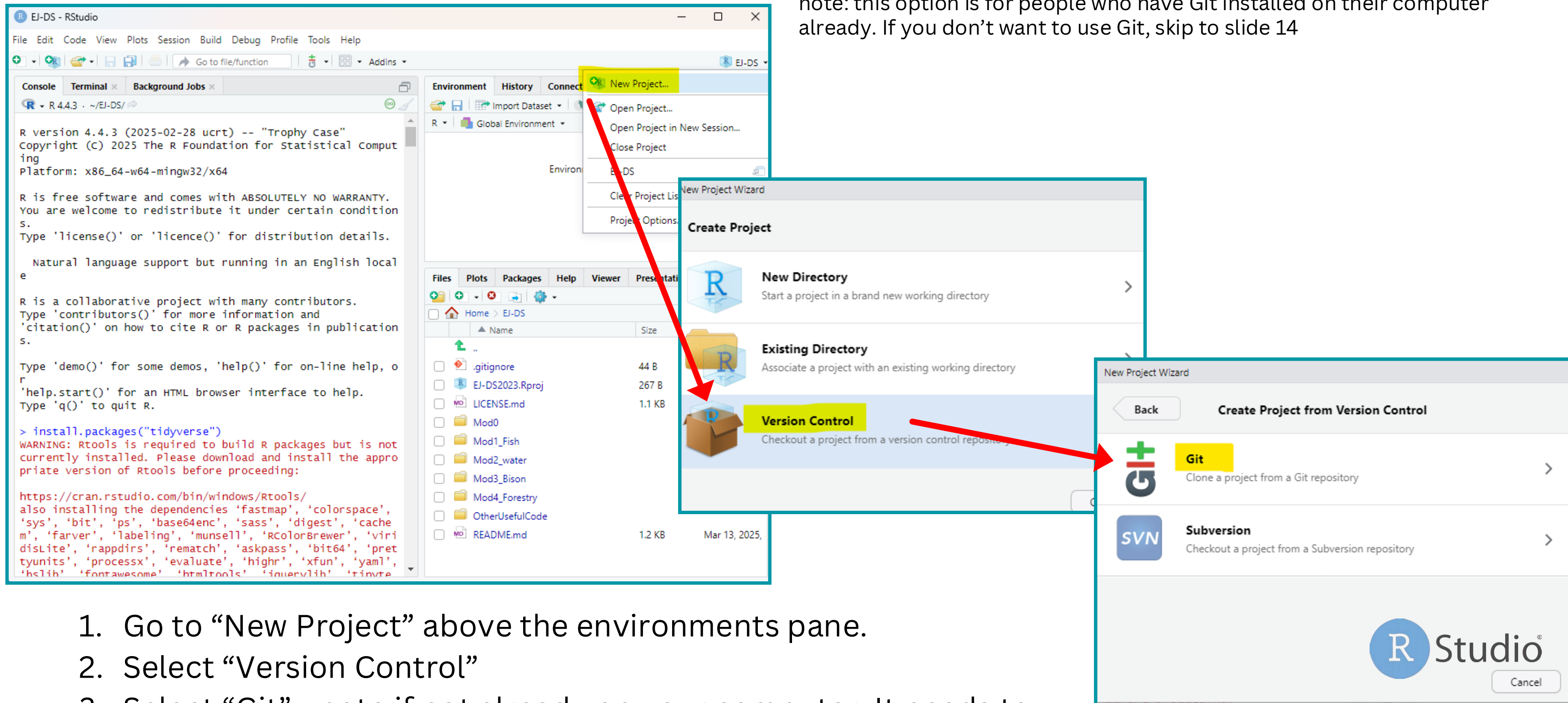
```
1 library(ggplot2)
2 mpg_plot <- ggplot(mpg, aes(x = displ, y = hwy)) +
3   geom_point(aes(colour = class))
4
5 mpg_plot
6
```
- Environments:** Located at the top right, it shows the `Global Environment` with a table of objects:

Name	Type	Len...	Size	Value
mpg_plot	gg	9	29.1...	List of 9
- Console:** Located at the bottom left, it shows the execution of the R script:

```
> library(ggplot2)
> mpg_plot <- ggplot(mpg, aes(x = displ, y = hwy)) +
+   geom_point(aes(colour = class))
>
> mpg_plot
>
```
- Output:** Located at the bottom right, it displays a scatter plot of `hwy` vs `displ` colored by `class`. The legend shows the following classes: 2seater, compact, midsize, minivan, pickup, subcompact, and suv.

Create a project in R Studio and connect files from GitHub

note: this option is for people who have Git installed on their computer already. If you don't want to use Git, skip to slide 14



The screenshot illustrates the process of creating a new R project in R Studio and connecting it to a version control system. The main window shows the R console with the R version 4.4.3 and the 'Trophy Case' copyright notice. The 'Environment' pane is open, showing the 'Global Environment'. The 'Files' pane displays the project structure, including files like .gitignore, EJ-DS2023.Rproj, LICENSE.md, and various Mod folders. The 'New Project Wizard' dialog is open, showing the 'Create Project' section. The 'Version Control' option is selected, and the 'Git' option is chosen under 'Create Project from Version Control'. The 'R Studio' logo is visible in the bottom right corner.

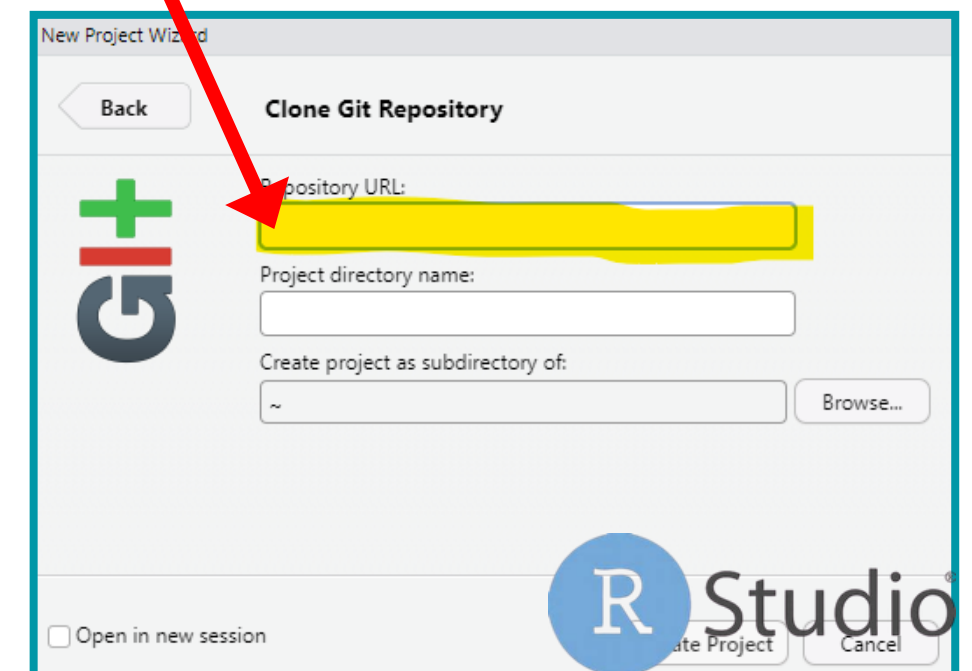
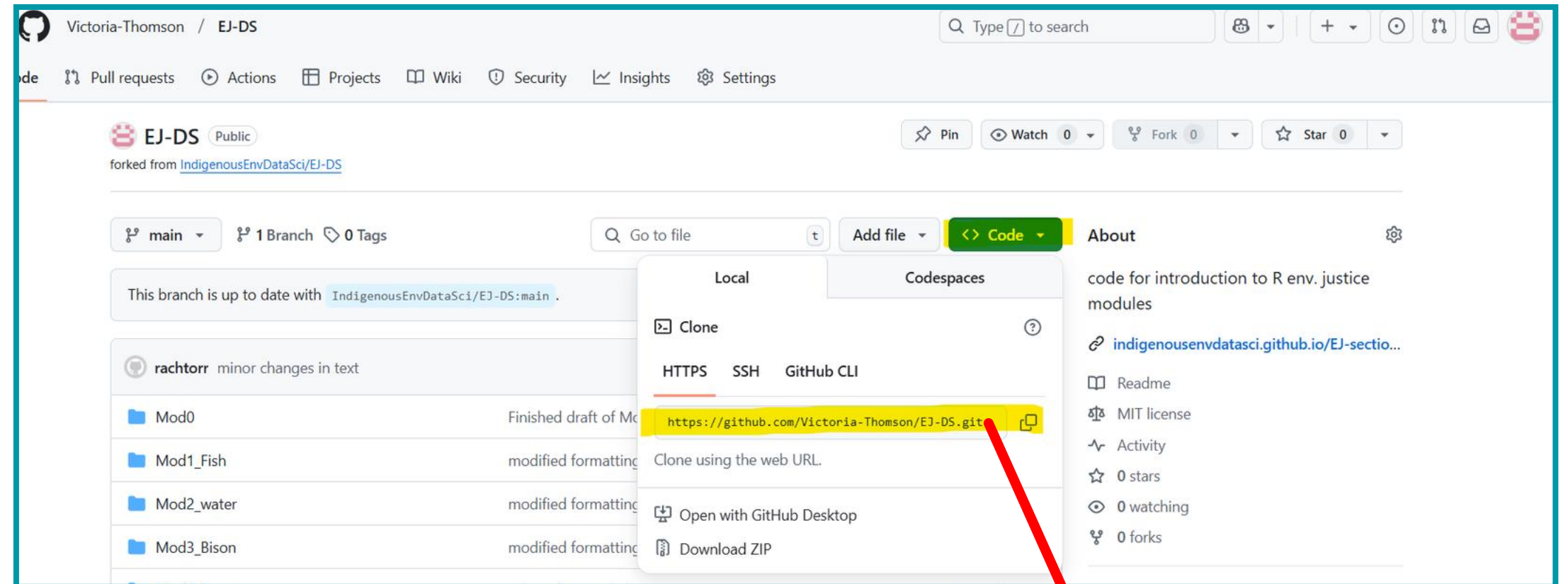
1. Go to “New Project” above the environments pane.

2. Select “Version Control”

3. Select “Git” - note if not already on your computer, It needs to be [downloaded](#)

Create a project in R Studio and connect files from GitHub

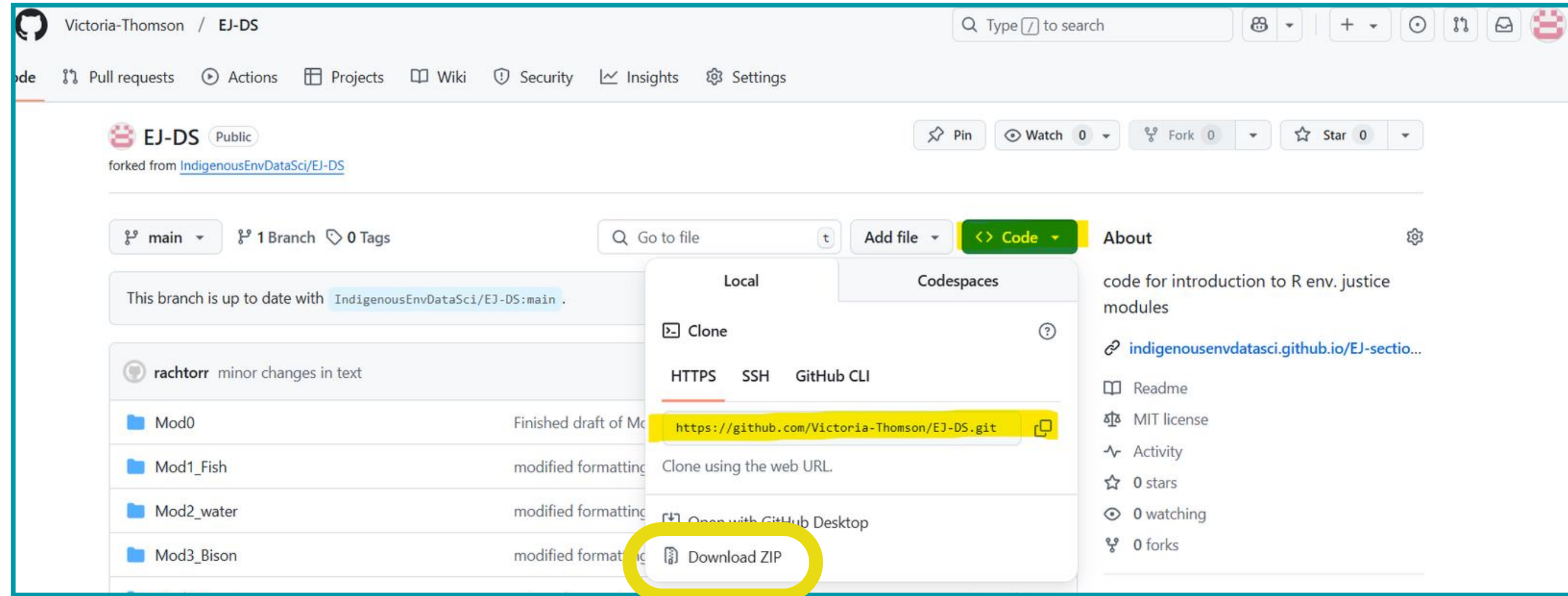
1. Select “code”
2. Copy the URL from Github and paste it into the “Repository URL”



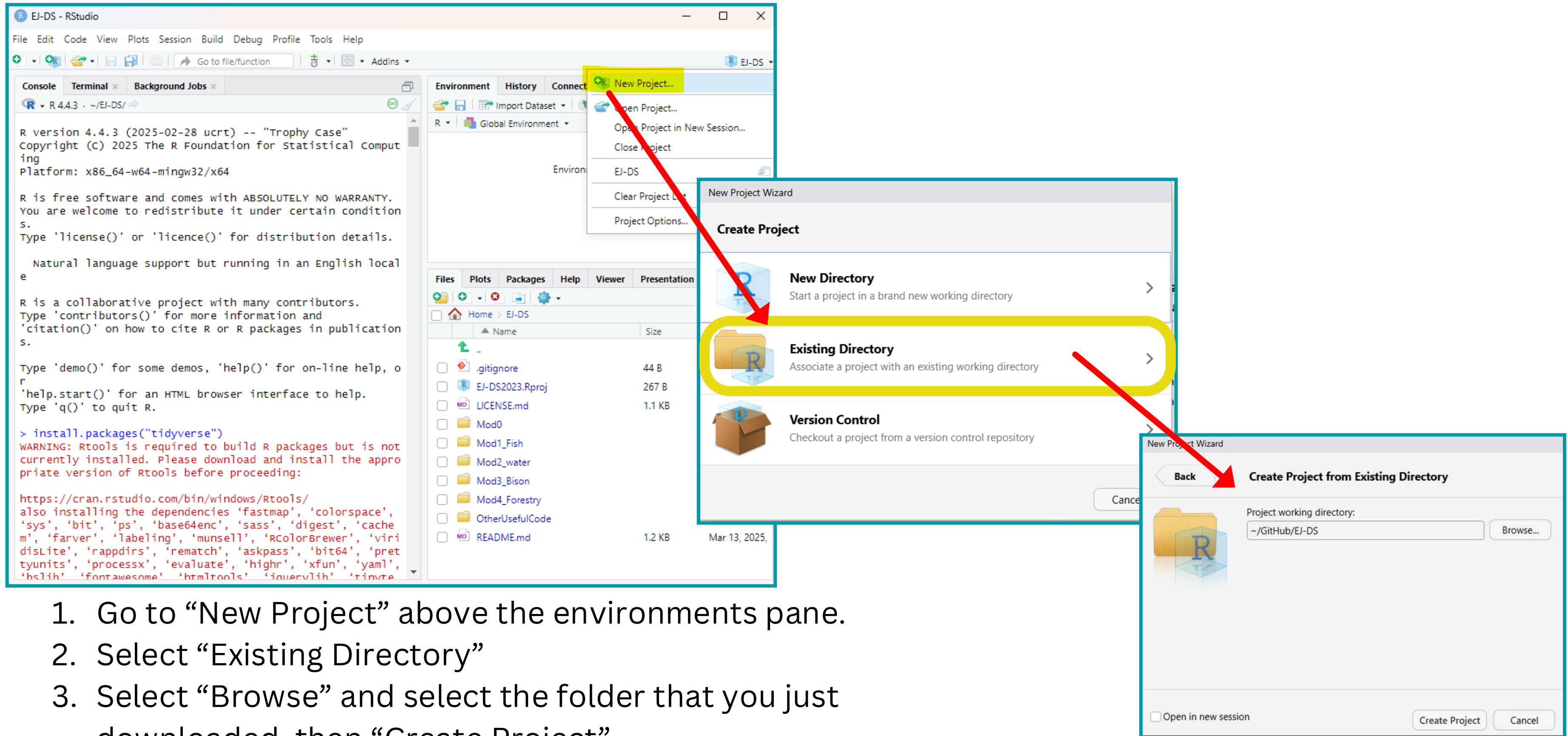
Create a project in R Studio with downloaded ZIP folder

note: this option is if you don't have Git installed on your computer. If you already created a project with Git, skip to slide 16

1. Select “code”
2. Download Zip file
3. Extract folder from Zip file and save in a familiar location



Create a project in R Studio and connect files from folder



The screenshot illustrates the steps to create a new R project in R Studio. The main window shows the R console with the R version 4.4.3 and the 'Trophy Case' copyright notice. The 'Environment' pane is visible, showing the 'Global Environment'. The 'Files' pane displays the contents of the 'EJ-DS' directory, including files like '.gitignore', 'EJ-DS2023.Rproj', 'LICENSE.md', 'Mod0', 'Mod1_Fish', 'Mod2_water', 'Mod3_Bison', 'Mod4_Forestry', 'OtherUsefulCode', and 'README.md'. The 'New Project' menu is open, showing options like 'New Project...', 'Open Project...', 'Open Project in New Session...', 'Close Project', 'EJ-DS', 'Clear Project List', and 'Project Options...'. The 'New Project Wizard' dialog is shown, with the 'Existing Directory' option selected. The 'Create Project from Existing Directory' sub-dialog is also visible, showing the 'Project working directory' as '~/GitHub/EJ-DS'.

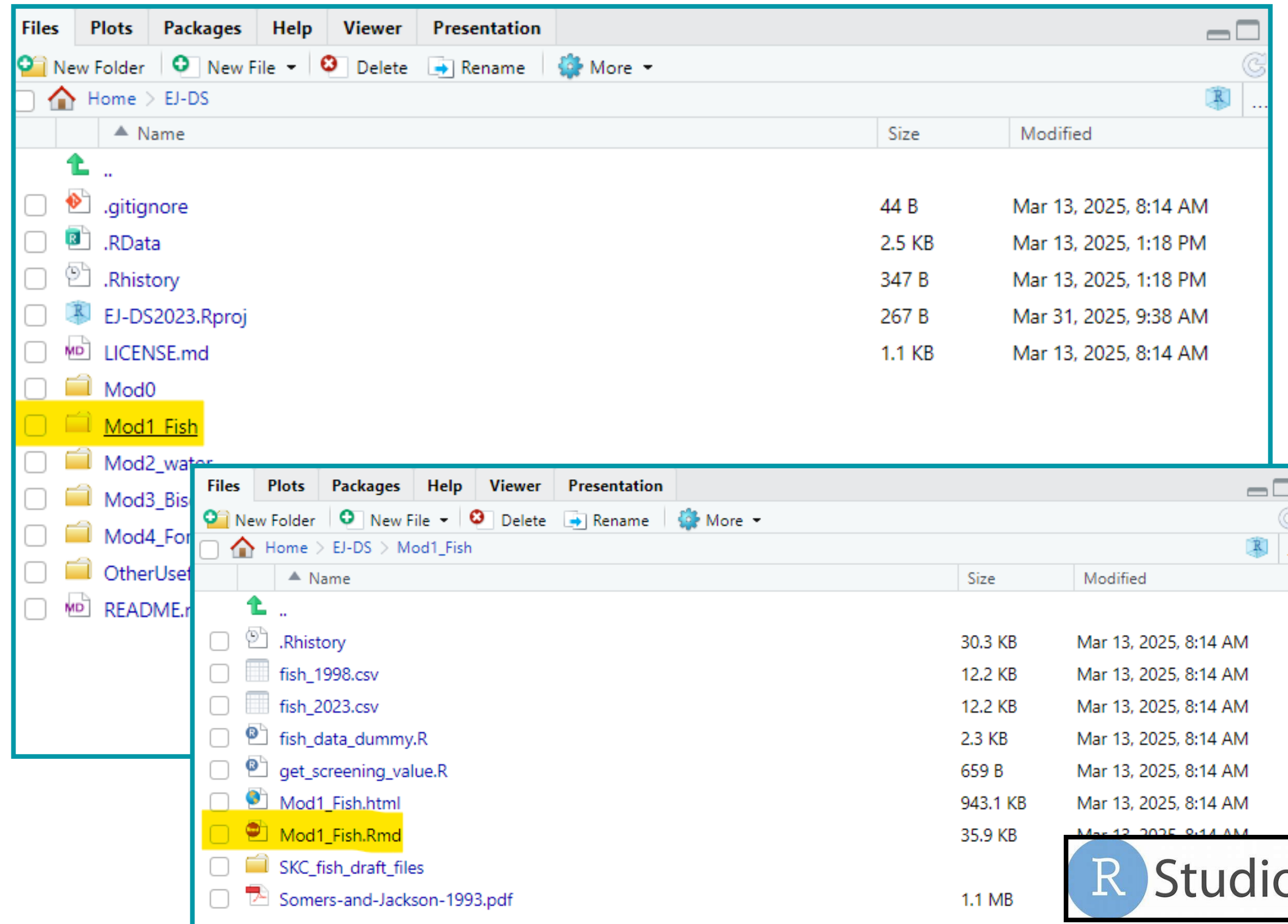
1. Go to “New Project” above the environments pane.

2. Select “Existing Directory”

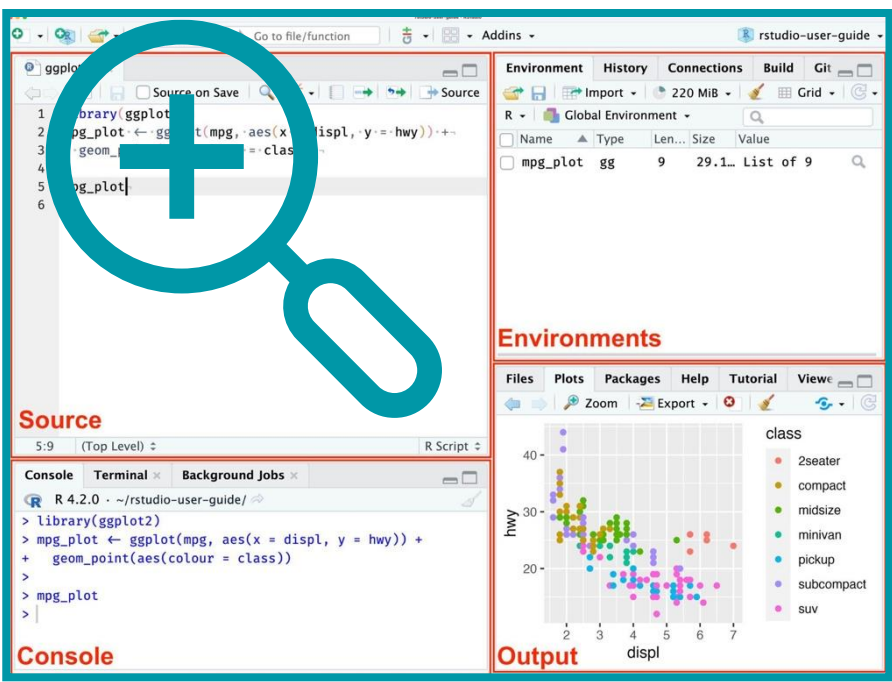
3. Select “Browse” and select the folder that you just downloaded, then “Create Project”

Access the R markdown file for Module 1

1. In the output console, select Mod1_Fish folder.
2. Then, select the file called “Mod1_Fish.Rmd”
 - a. Rmd stands for R markdown



Familiarize yourself with information in the source editor



Run code: press “play” to run the code chunk

Signals beginning of code chunk to R

Code chunk: an area with grey background that contains code

```
31 {r}
32 # In R, if you type the name of a data set, R will print it out.
33 # You can see that by running this code chunk, which will print
34 # a description of "pressure", which is a "data frame"
35 # with 19 rows and 2 columns. df[19,2]
36 # Then R will show you the first few lines of that data.
37
38 # If you, instead, type "pressure" in the Console window, below,
39 # R will print the whole dataframe in that window.
40
41 # The line below is the only line of code in this chunk.
42 # the rest is all comments.
43
44 pressure
45 }
```

Comments:
anything following a pound sign “#” within a grey code chunk; R does NOT run this code

Signals end of code chunk to



Read the Introduction code in the source editor

2 The code then shows up here!

```
1 ---
2 title: "Module 1: Fish and heavy metal contaminants"
3 output: html_document
4 date: "updated 2024-7-29"
5 ---
6 {r setup, include=FALSE}
7 knitr::opts_chunk$set(echo = TRUE)
8
9
10 # Introduction
11
12 Members of the K'avi have fished the waters on their land for many years to provide food
13 for their community. To manage these waters, K'avi tribal fishery managers have been
14 monitoring fish growth rates across the local water bodies in their community.
15
16 However, many members of the community have begun to experience health problems associated
17 with heavy metal poisoning.
18 Historically, a factory used to be located upstream of the riverways fished by the K'avi,
19 and waste from this factory was dumped into the nearby waters. Tribal fishery managers are
20 now concerned that members of the community
21 are being exposed to heavy metals through the fish that have been caught in these waters.
```

1 Select the Rmd file.

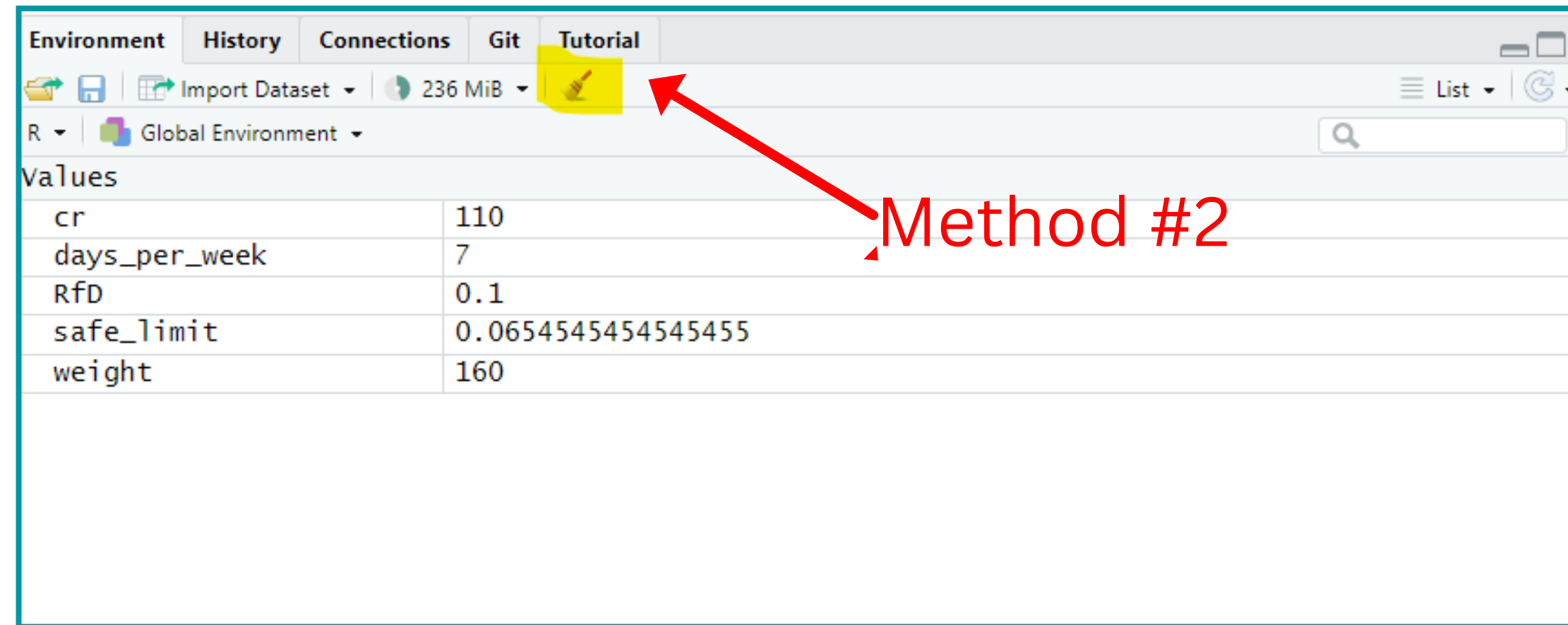
Name	Size	Modified
..		
.Rhistory	30.3 KB	Mar 13, 2025, 8:14 AM
fish_1998.csv	12.2 KB	Mar 13, 2025, 8:14 AM
fish_2023.csv	12.2 KB	Mar 13, 2025, 8:14 AM
fish_data_dummy.R	2.3 KB	Mar 13, 2025, 8:14 AM
get_screening_value.R	659 B	Mar 13, 2025, 8:14 AM
Mod1_Fish.html	948.1 KB	Mar 13, 2025, 8:14 AM
Mod1_Fish.Rmd	38.9 KB	Mar 13, 2025, 8:14 AM
SKC_fish_draft_files		
Somers-and-Jackson-1993.pdf	1.1 MB	Mar 13, 2025, 8:14 AM

R Studio

Clear the environment

You can do this in one of two ways:

1. In the source console, run the code chunk (press the green play icon),
OR
2. In the environments console, click on the broom.



```
66 - ### Clearing Environment
```

```
67
```

```
68 The first thing we do anytime we start working in R is cleaning our environment. Your environment is  
where R stores all of the objects it is working with since the last time you cleared your environment.  
We clear the environment each time we open R so that there isn't anything still open from the last time  
you were using R that will mess up what we are trying to do this session. You can clear your  
environment with the following code or by clicking the broom icon that can normally be found on the top  
right section of RStudio.
```

```
69
```

```
70 **NOTE TO SKC/ND DEVELOPERS:**
```

```
71 **THIS AND THE SECTION ON LOADING PACKAGES WILL ALSO BE COVERED IN MODULE 0**
```

```
72
```

```
73 ~~~{r}
```

```
74 rm(list = ls())
```

```
75 ~~~
```

```
76
```

```
77
```

Method #1



Install the tidyverse Package

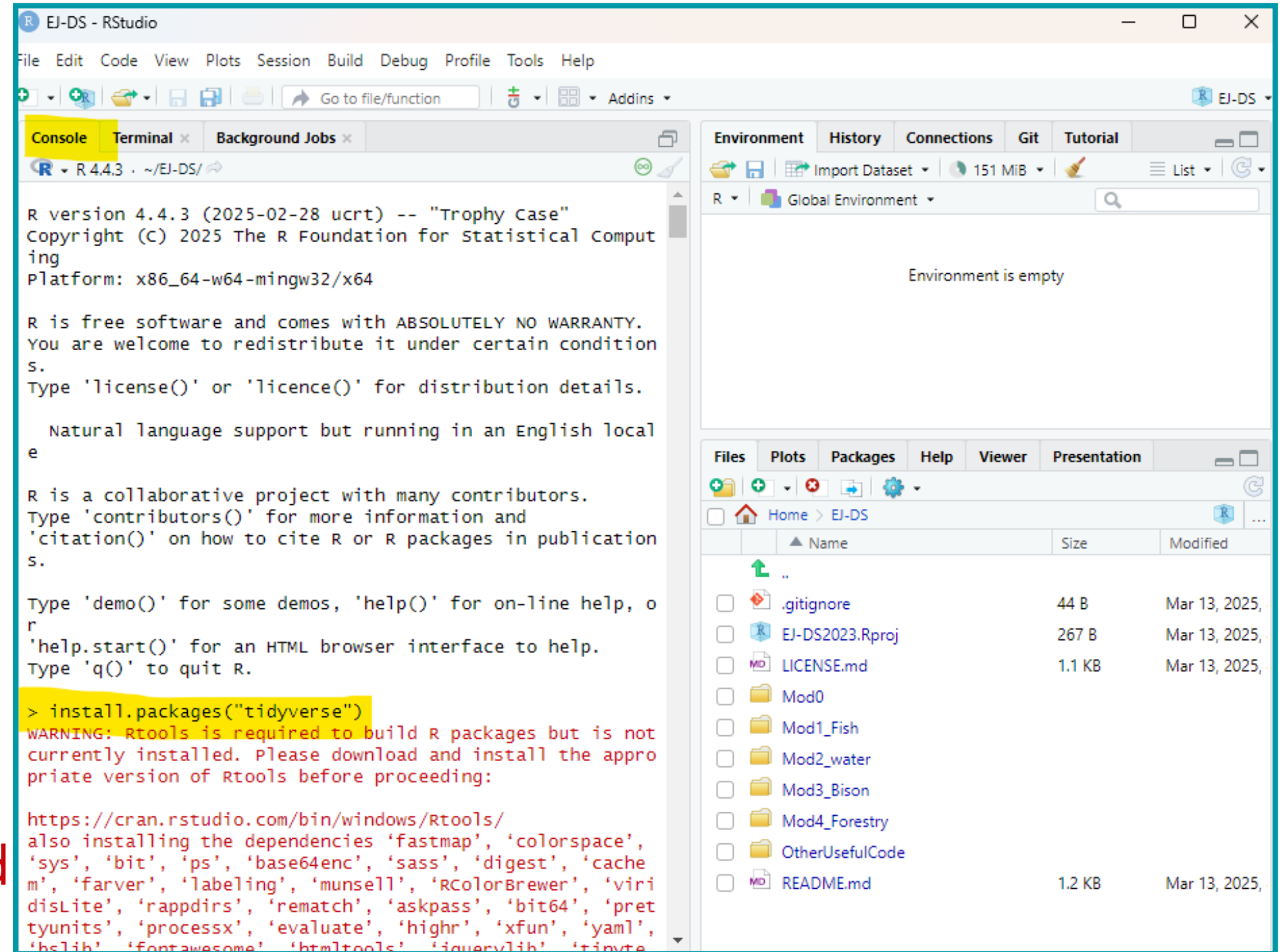
What is a **package**?

"Packages" are computer code that someone wrote to do a particular thing. By loading a package, R downloads that code, so that you can do that thing yourself.

Type the following in the console:

`install.packages("tidyverse")`

Note: I got a Warning that I should install Rtools to build r packages...you don't need to download Rtools



```
R version 4.4.3 (2025-02-28 ucrt) -- "Trophy Case"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> install.packages("tidyverse")
WARNING: Rtools is required to build R packages but is not
currently installed. Please download and install the appropriate
version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
also installing the dependencies 'fastmap', 'colorspace',
'sys', 'bit', 'ps', 'base64enc', 'sass', 'digest', 'cachem',
'farver', 'labeling', 'munsell', 'RColorBrewer', 'viridisLite',
'rappdirs', 'rematch', 'askpass', 'bit64', 'prettyunits',
'processx', 'evaluate', 'highr', 'xfun', 'yaml', 'bslib',
'fontawesome', 'htmltools', 'jquerylib', 'tinytex'
```

Load the tidyverse Package

Type the following into the console:

`library(tidyverse)`

```
> library(tidyverse)
── Attaching core tidyverse packages ─────────────────── tidyverse 2.0.0 ──
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr   1.5.1
✓ ggplot2     3.5.1    ✓ tibble     3.2.1
✓ lubridate   1.9.4    ✓ tidyr      1.3.1
✓ purrr       1.0.4
── Conflicts ─────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
Use the conflicted package to force all conflicts to become errors
>
```

Note about packages

Note: You only need to **install** a package once.

You DO need to run **library** of the packages of interest at the beginning of every session.



What does the package do? What functions are we learning?

Package	What does it do?	What functions are we using?
tidyverse	used for 'tidying' data: wrangling, analysis, visualizing	select(), filter(), mutate(), ggplot(), summarize()
ggplot2	Makes plots	aes(), geom_jitter(), geom_point(), geom_hline()